

HW1

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Status

Q1: Started Q2: Done (Minor Touch-ups) Q3: Q4: Q5: Q6: Q7: Q8: Done

Q1: A.3

Let $I = [0, 1]$, $\Omega = \mathbb{R}$ and for $\alpha \in \mathbb{R}$, $A_\alpha = (\alpha - 1, \alpha + 1)$, the open interval $\{x : \alpha - 1 < x < \alpha + 1\}$.

(a)

Show that $\bigcup_{\alpha \in I} A_\alpha = (-1, 2)$ and $\bigcap_{\alpha \in I} A_\alpha = (0, 1)$.

Answer

To start, note that $A_0 = (-1, 1)$ and $A_1 = (0, 2)$.

(b)

Suppose $J = \{x : x \in I, x \text{ is rational}\}$. Find $\bigcup_{x \in J} A_x$ and $\bigcap_{x \in J} A_x$.

Answer

Q2: A.5

Show that $X \equiv \{0, 1\}^{\mathbb{N}}$, the set of all sequences $\{\omega_i\}_{i \in \mathbb{N}}$ where each $\omega_i \in \{0, 1\}$, is **uncountable**. Conclude that $\mathcal{P}(\mathbb{N})$ is uncountable.

Answer

A.5 Solution

Sketch: To prove $X \equiv \{0, 1\}^{\mathbb{N}}$ is uncountable, and consequently $\mathcal{P}(\mathbb{N})$ is uncountable, we must first establish (1): X is uncountable and then (2): $\mathcal{P}(\mathbb{N})$ and X are in bijection.

(1): X is uncountable

Suppose, for contradiction, that X is countable. Since X is infinite (it contains, e.g., one sequence for every finite subset of $\mathbb{N}$), countability would mean there exists a surjection $h : \mathbb{N} \rightarrow X$ enumerating every element of X : $h(1), h(2), h(3), \dots$

Write $h(n) = (h(n)_1, h(n)_2, h(n)_3, \dots)$ for the n -th sequence in the list, with $h(n)_i \in \{0, 1\}$ its i -th coordinate.

Define $\delta = (\delta_i)_{i \in \mathbb{N}}$ by flipping the diagonal entries: $\delta_i = 1 - h(i)_i$, $i \in \mathbb{N}$.

Then $\delta \in X$, since each $\delta_i \in \{0, 1\}$. But for every $n \in \mathbb{N}$, $\delta_n = 1 - h(n)_n \neq h(n)_n$, so δ and $h(n)$ disagree in coordinate n , i.e. $\delta \neq h(n)$. This holds for **every** n , so $\delta \notin \{h(1), h(2), h(3), \dots\}$, contradicting the assumption that h is surjective (enumerates all of X).

Hence no such enumeration exists, and X is uncountable.

(2): $\mathcal{P}(\mathbb{N})$ and X are in bijection

Define $\Phi : \mathcal{P}(\mathbb{N}) \rightarrow X$ by

$$\Phi(A) = \{\omega_i\}_{i \in \mathbb{N}}, \quad \omega_i = \begin{cases} 1 & \text{if } i \in A \\ 0 & \text{otherwise} \end{cases}, \quad \forall A \in \mathcal{P}(\mathbb{N}),$$

i.e. $\Phi(A)$ is the indicator sequence of A : coordinate i is 1 if $i \in A$ and 0 otherwise. We show Φ is a bijection.

- **Injective:** if $A \neq B$, then without loss of generality there is some $i \in A \setminus B$, so $\Phi(A)_i = 1 \neq 0 = \Phi(B)_i$, hence $\Phi(A) \neq \Phi(B)$.
- **Surjective:** given any $\omega = (\omega_i)_{i \in \mathbb{N}} \in X$, let $A = \{i \in \mathbb{N} : \omega_i = 1\} \in \mathcal{P}(\mathbb{N})$. Then $\Phi(A) = \omega$ by construction.

Hence Φ is a bijection, so $|\mathcal{P}(\mathbb{N})| = |X|$.

Conclusion

Bijections preserve cardinality, so $|\mathcal{P}(\mathbb{N})| = |X|$. By Step 1, X is uncountable. Therefore $\mathcal{P}(\mathbb{N})$ is uncountable.

Remarks

This recovers the general fact $|\Omega| < |\mathcal{P}(\Omega)|$ (proved for $\Omega = \mathbb{N}$ directly in the first proof) as a special case. That argument builds, for any purported bijection $g : \mathbb{N} \rightarrow \mathcal{P}(\mathbb{N})$, the set $A = \{i \in \mathbb{N} : i \notin g(i)\}$, which lies in no $g(i)$: a preimage x with $g(x) = A$ would force $x \in A \iff x \notin g(x) = A$, a contradiction. Translating through Φ , this A is exactly the diagonal sequence δ constructed above, since $\delta_i = 1 - \Phi(g(i))_i$ says precisely $\delta_i = 1 \iff i \notin g(i)$. The two diagonalizations are the same argument, one on sequences and one on sets.

Q3: A.9

Apply the principle of induction to establish the following:

(a)

$$\text{For each } n \in \mathbb{N}, \sum_{j=1}^n j^2 = \frac{n(n+1)(2n+1)}{6}.$$

Answer

(b)

For each $n \in \mathbb{N}$, $x_1, x_2, \dots, x_k \in \mathbb{R}$,

(i)

$$(\text{The binomial formula}). (x_1 + x_2)^n = \sum_{r=0}^n \binom{n}{r} x_1^r x_2^{n-r}.$$

Answer

(ii)

(The multinomial formula).

$$(x_1 + x_2 + \dots + x_k)^n = \sum \frac{n!}{r_1! r_2! \dots r_k!} x_1^{r_1} x_2^{r_2} \dots x_k^{r_k},$$

where the summation extends over all $(r_1, r_2, \dots, r_k)$ such that $r_i \in \mathbb{N}$, $0 \leq r_i \leq n$, $\sum_{i=1}^k r_i = n$.

Answer

Q4: A.11

Show that the set $A = \{r : r \in \mathbb{Q}, r^2 < 2\}$ is bounded above in $\mathbb{Q}$ but has no l.u.b. in $\mathbb{Q}$.

Answer

Q5: A.12

Show that for any two sequences $\{x_n\}_{n \geq 1}, \{y_n\}_{n \geq 1} \subset \mathbb{R}$,

$$\begin{aligned} \lim_{n \rightarrow \infty} x_n + \lim_{n \rightarrow \infty} y_n &\leq \lim_{n \rightarrow \infty} (x_n + y_n) \leq \overline{\lim}_{n \rightarrow \infty} (x_n + y_n) \\ &\leq \overline{\lim}_{n \rightarrow \infty} x_n + \overline{\lim}_{n \rightarrow \infty} y_n. \end{aligned}$$

Answer

Q6: A.17

Find the radius of convergence, ρ , for the powers series $A(x) \equiv \sum_{n=0}^{\infty} a_n x^n$ where

(a)

$$a_n = \frac{n}{(n+1)}, \quad n \geq 0.$$

Answer

(b)

$$a_n = n^p, \quad n \geq 0, \quad p \in \mathbb{R}.$$

Answer

(c)

$$a_n = \frac{1}{n!}, \quad n \geq 0 \quad (\text{where } 0! = 1).$$

Answer

Q7: A.18

(a)

Find the Taylor series at $a = 0$ for the function $f(x) = \frac{1}{1-x}$ in $I \equiv (-1, +1)$ and show that it converges to $f(x)$ on I .

Answer

(b)

Find the Taylor series of $1 + x + x^2$ in $I = (1, 3)$, centered at 2.

Answer

(c)

Let

$$f(x) = \begin{cases} e^{-\frac{1}{x^2}} & \text{if } |x| < 1, x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

(i)

Show that f is infinitely differentiable at 0 and compute $f^{(j)}(0)$ for all $j \geq 1$.

Answer

(ii)

Show that the Taylor series at $a = 0$ converges but not to f on $(-1, 1)$.

Answer

Q8: A.24 (c)

Draw the open unit ball $B_p \equiv \{x : x \in \mathbb{R}^2, d_p(x, 0) < 1\}$ in $\mathbb{R}^2$ for $p = 1, 2$ and ∞ .

Answer

$p = 1, 2$ and ∞ are shown below in black, red, and blue respectively.

